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Task:

1. Create two threads to read two separate text files
 - 1.1 Same two files, try to read using the Task Async await.

Solution:

```
using System;
using System.IO;
using System.Threading;
```

```
class Program
{
    static void ReadFile1()
    {
        string filePath = "C:\\Users\\narendrakumar.s\\Documents\\myfile.txt";

        if (File.Exists(filePath))
        {
            string content = File.ReadAllText(filePath);
            Console.WriteLine($"Content : \\n{content}");
        }
        else
        {
            Console.WriteLine("File not found");
        }
    }

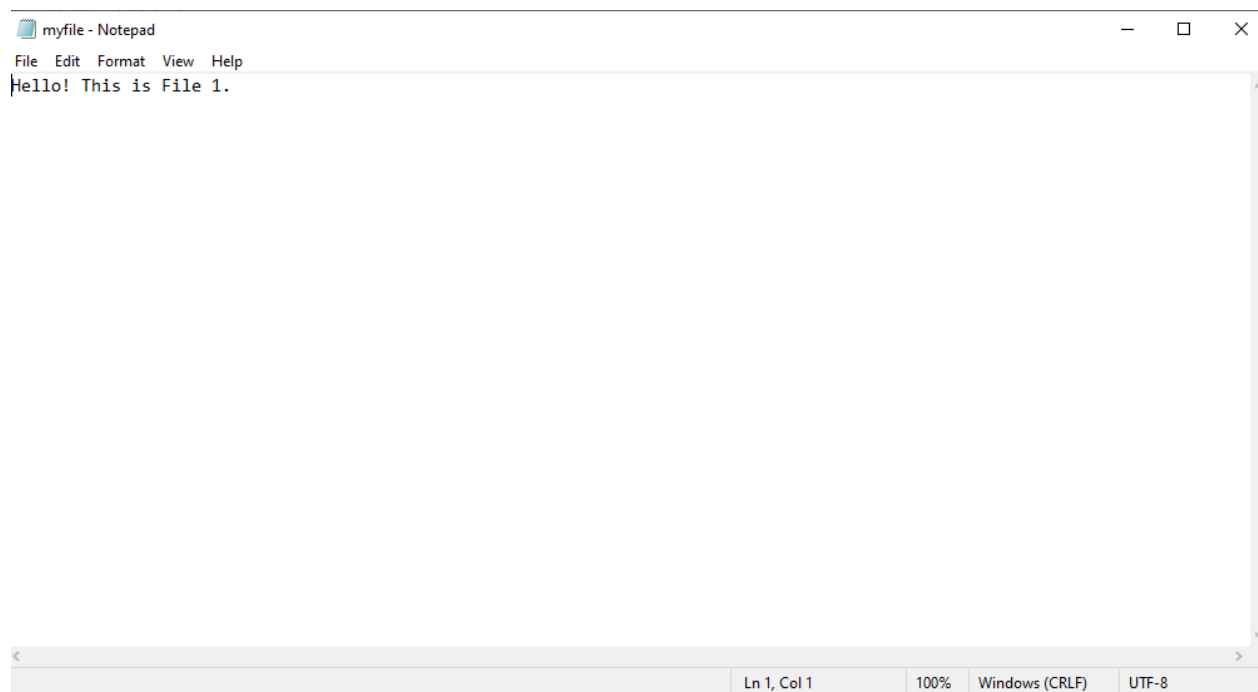
    static void ReadFile2()
    {
        string filePath = "C:\\Users\\narendrakumar.s\\Documents\\myfile2.txt";

        if (File.Exists(filePath))
        {
            string content = File.ReadAllText(filePath);
            Console.WriteLine($"Content is : \\n{content}");
        }
        else
        {
            Console.WriteLine($"{{filePath}} not found.");
        }
    }
}
```

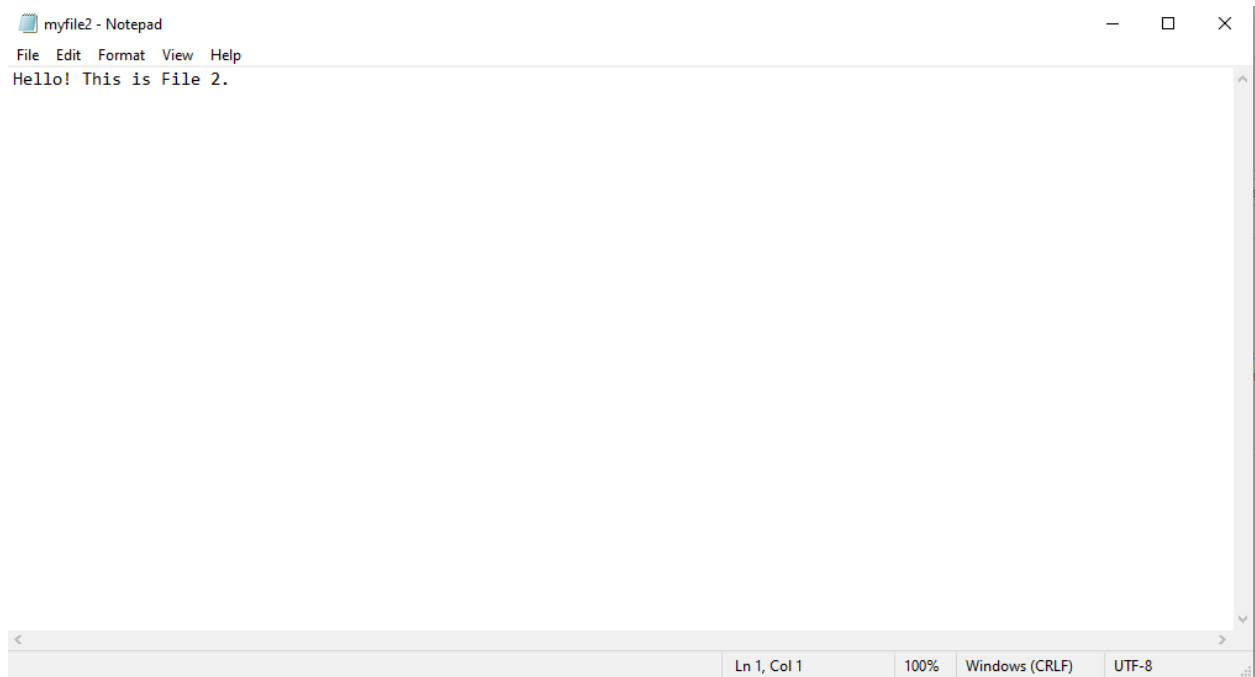
```
    }  
}  
  
static void Main()  
{  
    Thread t1 = new Thread(ReadFile1);  
    Thread t2 = new Thread(ReadFile2);  
  
    t1.Start();  
    Thread.Sleep(1000);  
    t2.Start();  
  
    t1.Join();  
    t2.Join();  
  
    Console.WriteLine("File Readed successfully using the Thread");  
}  
}
```

Output:

File 1:



File 2:



Console Output:



2. Create delegate use case between teacher class and student class:

- teacher method should have test_completed() method passed as delegate to student
- student class should have a method write_test() which will inturn call the parent delegate.

Solution:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Day4_HandSons_2
{
    internal class Teacher
    {
        public Student s;

        public Teacher()
        {
            s = new Student(test_done);
        }

        public void test_done(string m)
        {
            Console.WriteLine("Teacher Recevied: " + m);
        }

        public void start_test()
        {
            s.write_test();
        }
    }
}
```

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
```

```

namespace Day4_HandSons_2
{
    internal class Student
    {
        public delegate void mydel(string msg);
        public mydel d;

        public Student(mydel d1)
        {
            d = d1;
        }

        public void write_test()
        {
            Console.WriteLine("Student writing test");
            d("Test completed");
        }
    }
}

```

```

using System;

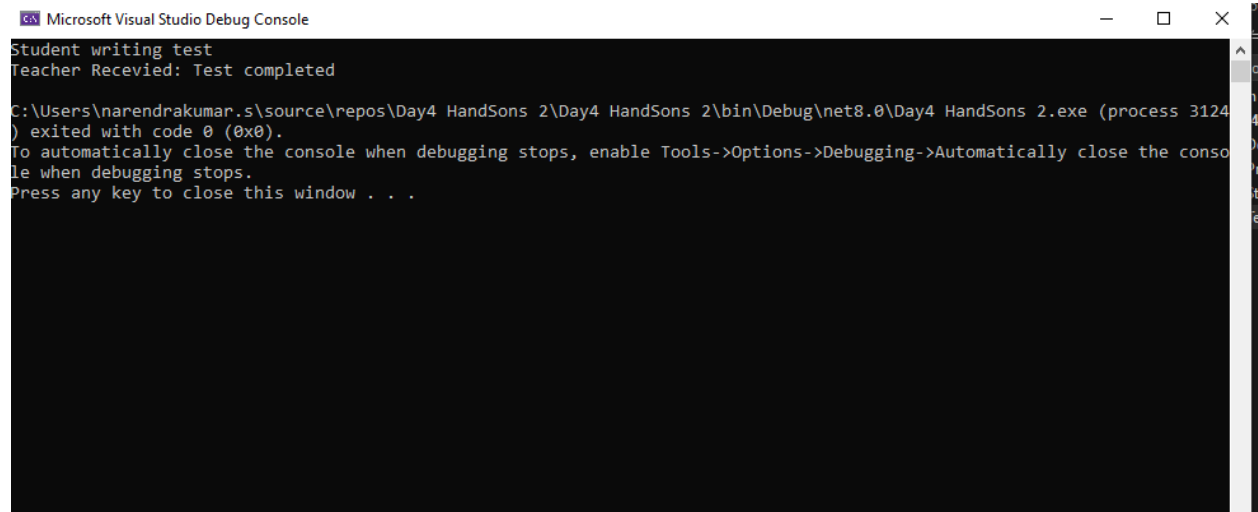
```

```

namespace Day4_HandSons_2
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Teacher t = new Teacher();
            t.start_test();
        }
    }
}

```

Output:



```
Microsoft Visual Studio Debug Console
Student writing test
Teacher Recevied: Test completed

C:\Users\narendrakumar.s\source\repos\Day4 HandSons 2\Day4 HandSons 2\bin\Debug\net8.0\Day4 HandSons 2.exe (process 3124) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```